



Study of indiscriminate distribution of restrained antimicrobial resistome of different environmental niches

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Abstract

Prophylactic usage and high persistent nature of several antibiotics have put selective pressure on the native microbial population that led to the emergence, propagation, and persistence of antibiotic resistance in nature. The surveillance of antibiotic resistome pattern and identification of points of intervention throughout the different environmental habitats will help to break the flow of antibiotic resistance from environmental bacteria to human pathogens. The present study compares the occurrence, diversity, and abundance of ARGs in industrial sludge, wetland sludge, and sediment sample contaminated with pharmaceutical discharge. Metagenomes were mined for the presence of ARGs against the ResFinder 3.2 database using BLASTn program. Pharmaceutical sample (2.52%) showed high degree of ARG abundance and richness as compared with ETP sludge (2.28%) and wetland sludge samples (1.29%). The modern resistome pattern represented by critically important resistance genes against tetracycline (*tetA*, *tetC*, *tetW*, *tetT*, and *tetS/M*) and quinolone (*qnrS*, *qnrVC*, and *qnrD*) was identified in pharmaceutical sediment sample. However, effluent treatment plant (ETP) sludge sample showed abundance of multidrug efflux pumps indicating the presence of primitive resistome profile. In conclusion, the indiscriminate distribution pattern of antibiotic resistance genes in three selected environmental sites suggests enrichment and distribution of environmental niche-driven resistance. The study also suggests effluent discharge site from pharmaceutical industries and ETPs as pivotal points of intervention for the mitigation of antibiotic resistance.

Keywords Antibiotic resistance gene · Metagenomics · Selective pressure · Resistome profile

Introduction

The world is entering a post-antibiotic era, and there is an urgent need of new antibiotics as resistance has shaped into a major global threat (Gebhard 2012). Several mechanisms to combat the lethal actions of antibiotics has evolved in microorganisms involving antibiotic inactivation or degradation by

enzymes, reduction in cell wall permeability, changes in antibiotic target sites, active efflux of antibiotic, and alternative metabolic pathways (Figure S1) (Blair et al. 2015). It has been propounded that antibiotic concentration below minimum inhibitory concentration (MIC) acts as Darwinian selectors (Forbes et al. 2017). Relatively, environmental persistence of various antibiotics such as ciprofloxacin (DT50:1153–3466 days), norfloxacin (DT50:1733 days) (Goulas et al. 2018), and sulphonamide (DT50:80 days) (Accinelli et al. 2007) leads to selection pressure and thus evolution in the indigenous microbial community to contain plethora of resistance genes (Munita and Arias 2016). Antibiotic resistance is becoming more vexed because of horizontal gene transfer (HGT) via transposons, integrons, plasmids, insertion sequences, and bacteriophages (Roberts 2011).

Large amounts of antibiotics manufactured and used in human and animal operations for disease prophylaxis and for growth promotion are released in partially metabolized forms in the wastewater treatment plants (WWTP) and sewage treatment plants (STP) (Deshmukh et al. 2016; Dafale et al.

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2020). Sewage and livestock manure often considered as a high risk factor for spreading antibiotic resistance has brought attention towards the solution for water pollution, and public health (de Araújo et al. 2020). Recent studies have recognized WWTPs and STPs as hotspots for breeding and dispersion of antibiotic-resistant bacteria (ARB) and antibiotic-resistant genes (ARG) in humans and animals (Yadav and Kapley 2019; Kraemer et al. 2019; Xia et al. 2019). Untreated sewage is rich in nutrients and cell density and has a range of antibiotics present at sub-MIC, which synergize to foster HGT events in existing bacteria (Bengtsson-Palme et al. 2018). An effective wastewater treatment is prerequisite before discharge into water bodies to circumvent the dispersion of ARGs and ARB into the environment (Barancheshme and Munir 2018). The biological and physicochemical treatment processes are proficient in a significant reduction in ARB but are not effective in ARG elimination. Currently, the guidelines for permitted levels of release of antibiotics or antibiotic resistance determinants into the environment are lacking (Pazda et al. 2019). The inefficiently treated effluents and bio-solids from WWTPs and STPs are received by surface waters (rivers, lakes, seas, or oceans) or into agricultural soils and crops (Xu et al. 2015; Zhang et al. 2009). At these polluted sites, the resistance is disseminated from autochthonous bacteria to commensals and pathogenic bacteria, making antibiotic resistance problem a public health concern (Fresia et al. 2019). Especially multidrug-resistant bacteria, including *Escherichia coli*, *Salmonella typhi*, and *Staphylococcus aureus*, which are causative agents for infectious diseases that have estimated to result in 70,000 deaths annually, are most dangerous (CDCP 2013; Laxminarayan et al. 2016; Xu et al. 2018). It is predicted to increase to 10 million deaths worldwide and 2 million in India annually by the year 2050 if current antibiotic resistance is not tackled (CDDEP 2015; Dixit et al. 2019).

Since the last few years, it has been noted that the trend of ARB and ARG studies is shifting from culturable approaches to the molecular and metagenomic approaches. The uncultured microbial communities inhabiting the WWTPs are known as the source of the ARGs transmitter (Turolla et al. 2018). The metagenomic approach can detect uncultured bacteria and do not suffer from the drawbacks of low-throughput and amplification bias. Hence, the use of a metagenomic approach to detect and screen broad-spectrum novel ARGs in WWTPs is encouraged. There is less understanding of origin of environmental resistance, its dispersal barriers, and dissemination routes. Studying antibiotic resistance origin and spread helps to generate reliable and comparable data for the monitoring of emergence of resistant pathogens, and for planning strategies to mitigate the threat of resistance. It entails the need for a coordinated, collaborative, and multidisciplinary initiative that operates at human, animal, agricultural, food, and environmental interfaces, termed as *One Health* approach (Dafale et al. 2020).

The aim of this study is to explore the resistance profile of highly persistent and extensively used antibiotics like sulphonamide, quinolone, tetracycline, and aminoglycoside, respectively, in the different ecological habitats. Samples of industrial and wetland sludge were selected for mining the occurrence, diversity, and abundance of ARGs and compared with pharmaceutical wastewater contaminated sediment using the metagenomic approach. This study focuses on sketching the flow to ARG contamination, understanding the structuring of dynamic resistome pattern and comparative resistome pattern of a particular habitat. The outcome highlights the need for framework towards prudent use of antibiotics and a robust treatment regime for combating environmental ARG pollution.

Materials and methods

Sample collection

The two activated sludge samples were collected, one from the effluent treatment plant (ETP) of industry (Mumbai, India) know for manufacturing of organic and inorganic chemicals. The second sample was collected from constructed wetland, Pandharkwada, near Nagpur (India), in the month of February 2019. The samples (10 L) were collected in a sterile container and transported to the lab in a 4 °C condition on the same day. The physicochemical characteristics (soluble COD, phosphate, and nitrogen) of both the collected samples were assayed with standard methods (APHA 1989, 2017). The metagenomic raw reads data of pharmaceutical discharge contaminated sediment (Hyderabad, India) was retrieved from MG-RAST (Accession number: mgm4701410.3) and was considered as a reference metagenome for comparison with the expectation that it has diverse resistome patterns.

Metagenomic DNA extraction

The activated sludge samples collected from both sites were centrifuged at 8000 rpm for 10 min, and 500 mg of the pellet was used for DNA extraction. Metagenomic DNA from two sludge samples was extracted using FastDNA Spin Kit for Soil (MP Biomedicals, USA) according to manufacturer's instructions. DNA quality and potential contamination were checked on 0.8% agarose gel. DNA concentration and purity were checked using Nanodrop DS-11+ (DeNovix) spectrophotometer. The concentrations of DNA extracted from ETP sludge and wetland sludge were 591.05 and 601.36 ng/μL. The "A260/280 ratio" of DNA from two samples ranged between 1.78 and 1.9.

Metagenome library construction sequencing and assembly

Metagenomic DNA of both ETP sludge and wetland sludge samples were sequenced using Illumina Nextseq pair-end platform (Figure S2). The sequencing library was constructed according to the manufacturer's instructions by using 1–2 µg of metagenomic DNA. Physical fractionation is applicable by Covaris Sonicator. Fractionation steps were checked by Agilent2100 and Q-PCR to ensure that sufficient enrichment of the target is achieved. Here a range of end repairing, A-tailing, ligation of sequencing adapters, size selection, and PCR enrichment steps was used to produce the library. Sequencing was performed using the Illumina platform after library clustering with paired-end reads. Clean data obtained from sequencing of ETP and wetland sludge sample were submitted to MG-RAST server for further functional studies (MG-RAST ID; mgm4858231.3 and mgm4858232.3). The pharmaceutical sediment metagenome (MG-RAST accession number: mgm4701410.3) sequenced by Illumina platform was selected as a reference metagenome for the study.

Bioinformatic analysis of metagenomic data

High throughput shotgun metagenome sequencing using Illumina NextSeq platform generated a total of 8 GB, 9 GB, and 11 GB data for the three samples ETP sludge, wetland sludge, and pharmaceutical sediments, respectively. The metagenomic data analysis statistics of individual samples were calculated by in-house Perl scripts (Table 1). Preliminary bioinformatics analysis of all three metagenomic samples was performed using MG-RAST server v4. Adapter trimming was done using a skewer to remove adapters, and

low-quality reads (Quality score < 20). Short reads were assembled into large contigs using the Bowtie 2 algorithm. FragGeneScan was used to predict ORFs in assembled contigs (Figure S2). Protein coding ORFs were annotated against COG database. Functional annotations were performed against Clusters of Orthologous Groups (COGs) database that comprises of 2091 orthologous groups of proteins. The parameters were configured to the maximum E value cut-off of e^{-5} and the minimum identity cut-off of 60%.

Relative ARG diversity analysis

A similarity and abundance of ARGs in selected metagenomes were done using ResFinder 3.2 database (Zankari et al. 2013). Local BLASTn (Linux release 2.7.1+ version) with E value cut-off of $\leq 1 \times 10^{-5}$ was run to annotate ARGs in query datasets against ResFinder database. Parameters for hit count in metagenome against ARG database were kept at a percent identity threshold of 90% and a minimum length of a nucleotide sequence that must overlap with the resistance gene to count as a hit at 60%. These parameters were chosen to minimize false-positive mapping in sample data sets. ResFinder database contains copious variants of the same gene. The read counts were aggregated to gene levels according to common gene names to circumvent unspecific mapping of reads to identical parts of homologous gene variants (Munk et al. 2017). A resistance gene count matrix was constructed from the DNA read mapping output, with genes (3097) and samples (3) constituting rows and columns, respectively. The output was represented in the form of a Stacked bar diagram using Microsoft Excel 2007, Venn Diagram using an online tool (<http://bioinformatics.psb.>

Table 1 Sequence details and data assembly statistics of selected metagenomes

MG RAST accession number	ETP sludge	Wetland sludge	Pharmaceutical sludge
Total number of bases uploaded	3,763,402,378 bp	4,030,108,981 bp	3,063,998,418 bp
Total number of sequences uploaded	2,37,53,451	2,41,44,299	30,336,618
Mean sequence length of bp uploaded	158 ± 30 bp	167 ± 40 bp	101 ± 0 bp
Mean GC content uploaded	63 ± 11 %	64 ± 12 %	56 ± 10 %
Artificial duplicate reads	28,17,863	45,07,561	7,169,147
Number of sequences failed QC	-	-	-
Total number of bases post-QC	3,297,554,108 bp	3,237,021,047 bp	2,087,791,798 bp
Total number of sequences post-QC	2,08,29,689	1,93,13,314	21,658,368
Mean sequence length post-QC	158 ± 30 bp	168 ± 41 bp	96 ± 11 bp
Mean GC count post-QC	63 ± 11 %	64 ± 12 %	55 ± 11 %
Predicted protein features	1,56,81,352	1,36,69,077	14,069,830
Predicted rRNA features	1,14,019	32,599	1, 809,369
Identified protein features	74,80,508	54,13,298	2,263,614
Identified rRNA features	13,042	4,697	7,476
Identified functional categories	Undefined	Undefined	1,596,054

[urgent.be/webtools/Venn/](http://www.urgent.be/webtools/Venn/)), and Heatmap using a web-tool ClustalViz (Metsalu and Vilo 2015).

The relative ARG diversity of each sample was calculated as a percentage of unique gene sequences of the ResFinder database found in each sample metagenomes (Mao et al. 2015).

Diversity(percentage)

$$= \frac{\text{Number of reads}}{\text{Number of genes in database}} \times 100$$

ARG abundance of each sample was calculated as parts per million (ppm) (Chen et al. 2016).

Abundance(parts per million)

$$= \frac{\text{Number of reads}}{\text{Total number of reads}} \times 10^6$$

Statistical analysis

One-way analysis of variance (ANOVA) and principal component analysis (PCA) was conducted using MINITAB version 17 (<https://www.minitab.com/en-us/products/minitab/>). The analysis of difference in the abundance and profiles of ARGs between the samples was done using ANOVA, whereas Tukey all-pairwise test was performed to make multiple comparisons. The $p < 0.05$ for confidence level at 95% was considered statistically significant. The PCA method was used to characterize and group the samples. All the three sequencing data sets were used as statistical cases, and antibiotic resistance gene types were used as variables ($n = 9$). PCA was performed using the coverage correlation matrix of the ARG types (Yang et al. 2013).

Results

Functional gene annotation and distribution analysis

The physico-chemical characteristic of ETP and constructed wetland samples were found to be 92 mg/l and 62 mg/l sCOD, 15.2 mg/l and 1.98 mg/l phosphate, and 38 mg/l and 15.45 mg/l bitrogen, respectively. All three metagenomes were sequenced on the Illumina platform to reduce annotation bias. Short reads (100 bp) obtained from the Illumina sequencing platform give less annotated read frequency but high read count, providing good depth for the gene calling (Sims et al. 2014). Antibiotic resistance genes annotated in the COG database were aminoglycoside N3'-acetyltransferase, beta-lactamase class C, cephalosporin hydroxylase, chloramphenicol 3-O-phosphotransferase, chloramphenicol O-acetyltransferase, glycopeptides resistance protein, streptomycin 6-

kinase, and vancomycin resistance protein. Selectively, the glycopeptides resistance gene was not found in pharmaceutical sediment, and ABC type antimicrobial peptide transport gene was not found in the wetland sludge sample (Fig. 1; Table S1).

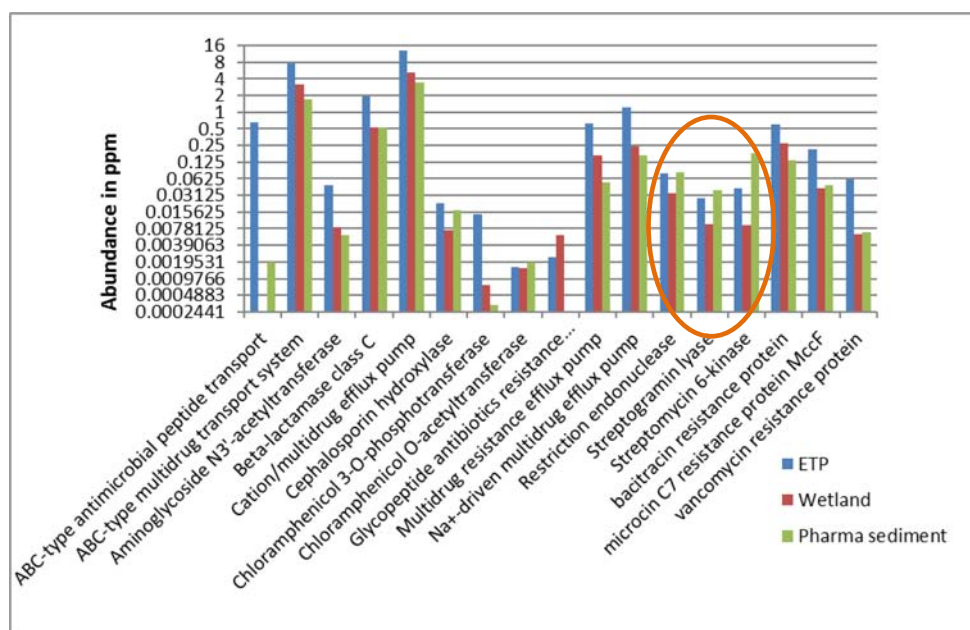
The analysis of bacterial defense mechanism genes in all metagenomes showed a relatively high occurrence of the cation/multidrug efflux pump genes, among others. Specifically, among efflux pumps, ABC-type peptide transport, ABC-type multidrug transport system, cation/multidrug efflux pump, multidrug resistance efflux pump, and Na⁺-driven multidrug efflux pump genes were identified in the highest amount in ETP sludge sample. The advance resistome pattern of pharmaceutical sediment was annotated with least amount of efflux pump genes, which belongs to primitive type of resistance genes. Genes for restriction endonuclease, streptogramin lyase, and streptomycin 6-kinase were found in higher abundance in pharmaceutical sediment sample as compare to ETP sludge and wetland sludge samples (Fig. 1).

Mining metagenomes to identify ARGs

Abundance and diversity of antibiotic resistance determinants

Collectively, 13 ARG types and 139 ARG subtypes were identified in all the three metagenomes. The total ARG abundance in pharmaceutical sediment (3794.98 ppm) was 89 times and 7.5 times higher than in the wetland sludge and ETP sludge samples, respectively (Table S2). As hypothesized, the diversity of ARGs in the pharmaceutical sediment sample (2.52%) was highest compared with that in ETP sludge (2.28%) and wetland sludge (1.29%) (Table S3; Fig. 2). However, strikingly, the ARG diversity of ETP sludge was also observed to be higher and relatively close to pharmaceutical sediment sample suggesting enrichment of ARGs in ETP sludge ($p = 0.161$) (Fig. 3). Quinolone resistance genes *qnrVC* (386.9 ppm), *qnrD* (335.9 ppm), and *qnrS* (71.6 ppm) were present in high amounts in pharmaceutical sediment sample, whereas *qnrB* (1.6 ppm) was present only in the ETP sludge sample. Nearly 42 variants of *qnrB* gene have been reported to occur, of which 23 variants were identified in the ETP sludge sample. Interestingly, the *qepA* gene was found in ETP sludge and wetland sludge samples and was absent from the pharmaceutical sediment sample. In contrast, *qnrD*, *qnrS*, and *qnrVC* genes were present only in the pharmaceutical sediment sample (Supplementary file 1). Analysis of tetracycline resistance (*tet*) genes identified 27 *tet* genes in the three metagenomes. Nearly 15 genes, namely, *tetA*, *tetC*, *tetW*, *tetT*, *tetS/M*, *tetO*, and *tetM* were present only in pharmaceutical sediment sample. Five *tet* genes that confer resistance via efflux mechanisms such as *tet1*, *tet42*, *tet33*, *tet43*, and *tetG* were found only in ETP sludge sample and *tet44*, *text*, and *tetX4* were only

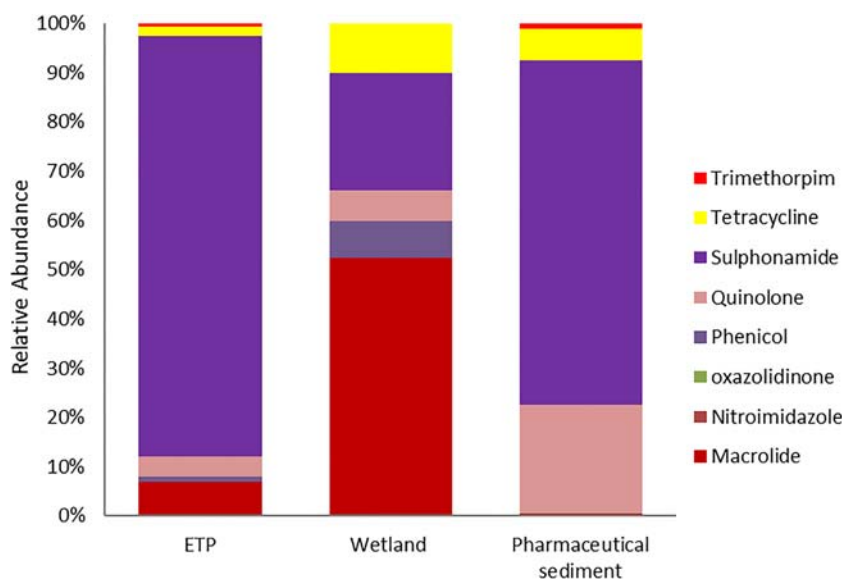
Fig. 1 Graphical representation of abundance profile of defence mechanism genes against COG database using MG-RAST in the selected environmental samples. Abundance of genes is presented in ppm (parts per million; one read in one million reads)



present in wetland sludge samples (Fig. 4a). Twenty-three macrolide resistance genes were annotated in the three metagenomes. Interestingly, genes conferring resistance via target modification and antibiotic inactivation mechanisms (*mph*, *ere*, and *erm*) were more prevalent in pharmaceutical sediment sample; however, the prevalence was not statistically significant ($p = 0.153$); viz, *mph(E)* and *ere(A)* genes were found in the pharmaceutical sediment sample in the highest amounts (*mph(E)*; 6.83 ppm, *ere(A)*; 5.22 ppm). Macrolide resistance genes that confer resistance through efflux mechanisms such as *srm(B)*, *car(A)*, *ole(C)*, and *ere(A)* genes were present in ETP sludge sample in higher amounts, whereas *ole(B)* gene was present in wetland sludge sample in highest amounts (*ole(B)*; 6.16 ppm) compared with that found in other

two samples (Fig. 4b). Nine phenicol resistance genes were annotated in the three metagenomes. ETP sludge and wetland sludge samples contained *floR_1* and *floR_2* genes in the highest amounts compared with other phenicol resistance genes and were absent in the pharmaceutical sediment sample. Resistance genes *catB1*, *catB7*, and *cml_1* were also absent in the pharmaceutical sediment sample. The pharmaceutical sediment sample contains *cmlA1* gene in the highest amounts and *catA1*, *catQ*, and *cml* genes were exclusively present in the only pharmaceutical sediment sample. *cmlA1* gene was the only phenicol resistance gene found in all the three samples (Fig. 4c; Table S2). Beta-lactam resistance genes were annotated only in wetland sludge and pharmaceutical sediment samples and completely absent from the ETP sludge sample.

Fig. 2 Comparison of ARG class abundance in metagenome data of representative environmental samples using Past3 software. Relative abundances of genes were summed to drug classes. Abundance of ARGs was normalized to the size of the sequencing data, and “ppm” refers to one ARGs-like tag in one million metagenomic sequencing tags



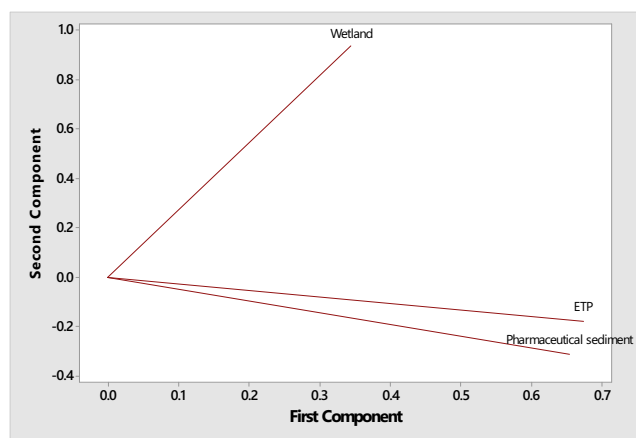


Fig. 3 PCA plot showing the ARG composition difference among ETP, Wetland and Pharmaceutical sediment samples

blaOXA-10 gene was the only beta-lactam resistance genes found in the wetland sludge sample, whereas it was absent in the pharmaceutical sediment sample. The pharmaceutical sediment sample had the presence of *blaNPS*, *blaCMY-4*, and *blaGES-2* genes, and these genes were not present in the wetland sludge sample (Fig. 4d). Rifampicin resistance genes (*arr-2*, *arr-3*, and *arr-6*) were present in all the three samples (Supplementary file 1). All three samples contain *arr-3* (rifampicin resistance gene) gene (ETP; 1.83, Wetland sludge; 0.57, pharmaceutical sediment; 93.54) in the highest amounts compared with *arr-2* and *arr-6* genes (Fig. 4e). Aminoglycoside resistance genes *aph(6)-Ia* and *aph(3)-Ib* were found in all the three samples. Pharmaceutical sludge samples do not contain *ant(3)-Ia* gene, which was present in only the wetland sludge sample. *aadA13*, *aadA7*, *aadA6*, and *aadA5* genes were exclusively present in only the pharmaceutical sediment sample (Fig. 4f). Only 3 sulphonamide resistance genes were annotated in the three samples. Sulphonamide resistance genes *sul1*, *sul2*, and *sul4* were present in high abundance in pharmaceutical sediment (2524.5 ppm), ETP sludge (423.81 ppm), and wetland sludge (9.98 ppm) samples ($p = 0.34$). The pharmaceutical sample contains *sul2* gene in the highest amounts (2203.58 ppm) compared with that found in the other two samples (ETP; 103.46, Wetland sludge; 1.45) (Fig. 4g). Wetland sludge sample does not contain *sul4* gene. Trimethoprim resistance genes were annotated only in pharmaceutical sediment and ETP sludge samples. Out of 13 trimethoprim resistance genes, only 3 resistance genes (*dfrA1*, *dfrA5*, and *dfrA7*) were mapped in the ETP sludge sample metagenome, and *dfrA1* gene was present in highest abundance compared with that in the pharmaceutical sediment sample. The pharmaceutical sediment sample contains 11 trimethoprim resistance genes except for *dfrA5* and *dfrA7* genes. The pharmaceutical sediment sample contains *dfrB3*, *dfrB5*, *dfrB6*, and *dfrB8* in the highest amounts among the mapped trimethoprim resistance genes.

Similarity analysis of ARG profiles

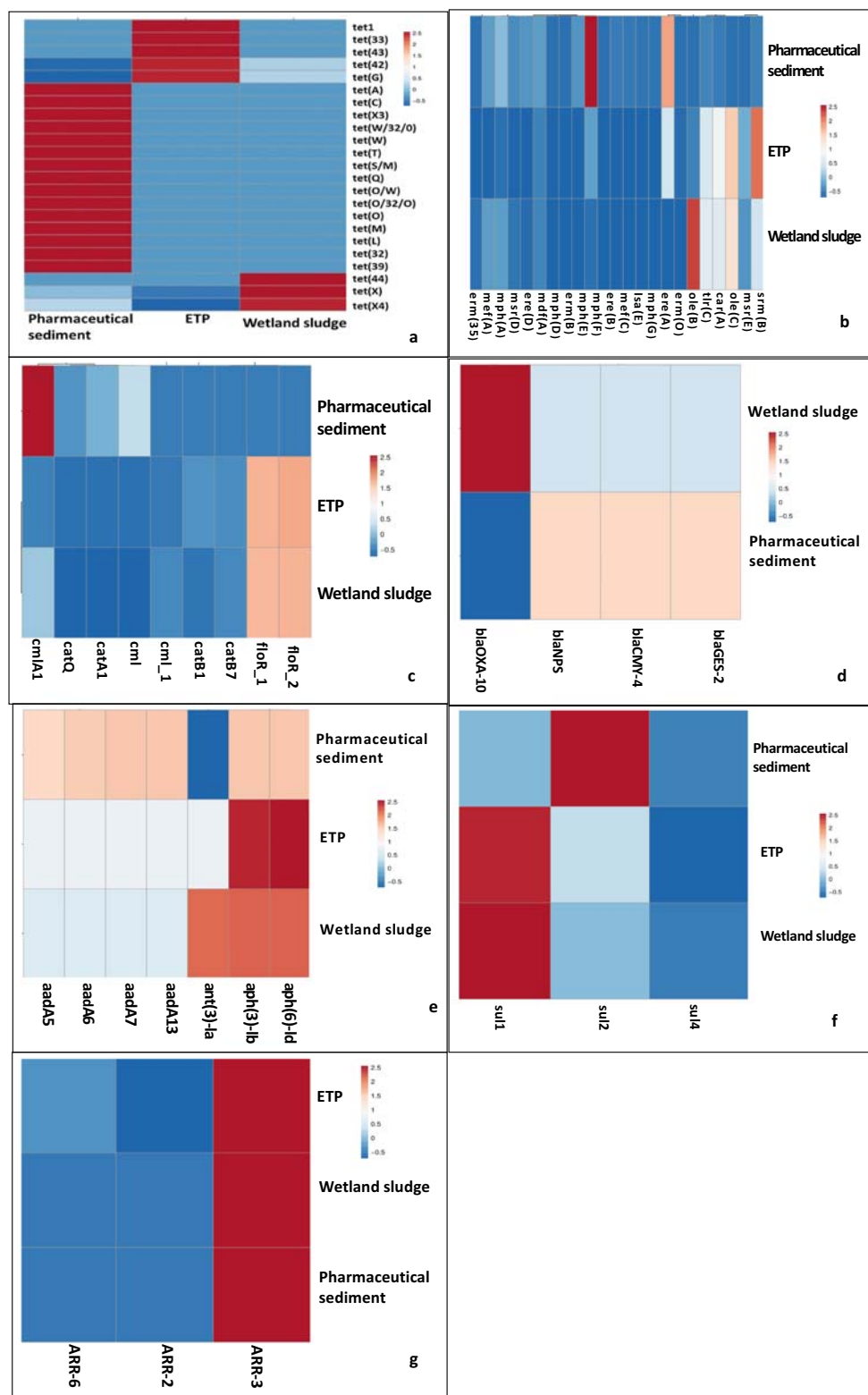
The 19 ARGs were shared among all the three metagenome samples, and these were basically the resistance genes for macrolide, tetracycline, sulphonamide, rifampicin, aminoglycoside, and phenicol (Fig. 5). As compared with the other two metagenome samples, pharmaceutical sediment sample has the highest degree of unique ARGs. Interestingly, our result has shown that the distribution profile of the ARGs depends on the habitat in the case of tetracycline and quinolone. The tetracycline resistance genes that were present only in the pharmaceutical sediment sample confer resistance through the target modification mechanism, whereas those present only in ETP sludge samples confer resistance through efflux pump mechanisms. The quinolone resistance genes present only in ETP sludge samples were conferring resistance through target modification mechanism, whereas the mechanism of those genes annotated in the pharmaceutical sediment sample is unspecified.

Discussion

The study aims to identify the occurrence, diversity, and abundance of ARGs in different environmental sampling sites, viz., industrial ETP sludge, wetland sludge, and pharmaceutical contaminated sediment. Antibiotic resistance genes were identified using ResFinder database, which has a collection of 3097 non-redundant nucleotide sequences known to convey antibiotic resistance phenotype to their host (Zankari et al. 2013). There are nearly 15 types of antibiotic resistance genes present in the ResFinder database. According to our results, the ETP sludge, wetland sludge, and pharmaceutical sediment metagenome samples showed a peculiar distribution pattern of resistance mechanism. ETP sludge has only *tet* efflux pumps in abundance and pharmaceutical sediment sample has target modification resistance genes in abundance. Most of the efflux pump genes and glycopeptides resistance protein gene were not found in pharmaceutical sediment, which confirms with the previous study of Li et al. (2015). The possible reason for such habitat bias in resistome profiles of pharmaceutical sediment samples might be due to the continuous selection pressure encountered by inhabiting microbial communities. This is further supported by our finding that ARG richness and high degree of abundance were observed in the pharmaceutical sample (2.52%) compared with that in ETP sludge (2.28%) and wetland sludge samples (1.29%). As for the distribution of ARG; the resistance genes for sulphonamide, quinolone, tetracycline, and aminoglycoside were abundant and commonly distributed than other ARGs types.

Resistance is mostly acquired by various mechanisms including decreased membrane permeability, acetylation of drugs (via *aac(6')-Ib* gene), inactivation of an antibiotic

Fig. 4 Abundance of antimicrobial resistance subclass of **a** tetracycline, **b** macrolide, **c** phenicol, **d** beta-lactam, **e** rifampicin, **f** aminoglycoside, and **g** sulphonamide. Colors represent log (ln) transformed relative abundances. Complete-linkage clustering of Pearson correlation coefficients was used to hierarchically cluster both samples and drug classes



(*aadA* gene, *tetX*) (Ogawara 2019), over expression of efflux pumps (*qepA* gene, *tetA*, *tetC*, *tet33b*) and transporter (*oqxAB* gene), ribosome protection (*tetO*), or by modification of DNA gyrase (topoisomerase II) and topoisomerase IV, thereby

interfering with replication, transcription, and translation (*qnr*-gene) (Poirel et al. 2012). Generally, efflux pumps are involved in primitive mechanisms of resistance and were chromosomally encoded (Nesme et al. 2014).

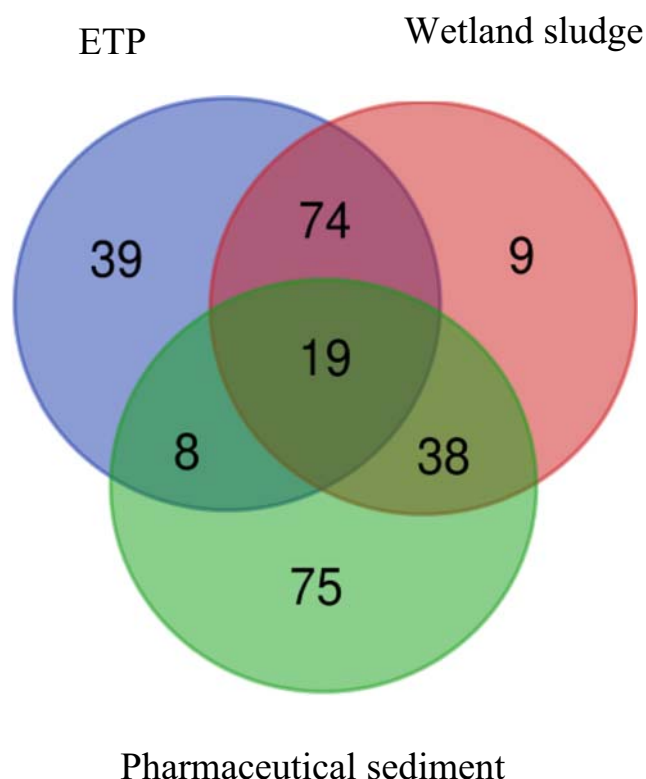


Fig. 5 Venn diagram showing common and unique ARG subtypes identified in ETP, wetland sludge, and pharmaceutical sediment

The result have shown that genes such as *aph(3')-Ib* and *aph(6)-Id* encode for phosphotransferases are found in all the three samples, while streptomycin 3' adenylyl transferase (*ant(3')-Ia*) is present in only the wetland sludge sample and aminoglycoside N-acetyltransferase (*aadA*) gene was found in only the pharmaceutical sediment sample. The *aph(3')-Ib* and *aph(6)-Id* were among most abundant antibiotic resistance genes in Indian river and was detected together with ISCR2 mobile genetic element (Bengtsson-Palme et al. 2014).

The macrolide resistance gene *erm* gene has 38 variants, of which 4 (*erm(35)*, *erm(B)*, *erm(F)*, *erm(O)*) were found in all the three sample metagenomes. Generally, erythromycin ribosomal methyltransferase (*erm*) gene confers resistance by methylating 23S ribosomal RNA at nucleotide A2058. Additionally, the efflux pump genes such as *srm(B)* gene was present at concentration of 9.27 ppm in ETP sample suggesting the prevalence of bacterial population harboring this gene in these sample as compared with other. Recently, a study by Bengtsson-Palme et al. (2019) reported that exposure of microbial community of industrial WWTPs to high levels of macrolide led to increased genetic mobility and thereby acquisition of resistance genes that are more commonly associated with MGE rather than for macrolide resistance. However, heavy metal especially copper and cadmium contamination along with macrolide contamination has shown to select for macrolide resistance genes in bacteria (Pal et al. 2015; Milaković et al. 2019).

The distribution of rifampicin resistance in samples highlighted interesting case. Pharmaceutical sediment sample has the highest of *arr* genes. Studies have reported the horizontal acquisition of *arr* genes (Baysarowich et al. 2008), mainly by clinical pathogens, whereas the environmental bacteria have mechanistic diversity for rifampicin resistance (Wright 2019). Such high amount specifically in one sample can be accounted for increased genetic mobility in bacterial population under selective pressure from antibiotics. In contrast, analysis showed that *floR* genes were not found in the pharmaceutical sediment sample. Generally, *floR* gene encodes for the efflux pump, which can effectively transport both chloramphenicol and florfenicol (Roberts and Schwarz 2016). Previously, studies have shown that chloramphenicol resistance genes (*floR* and *cml*) are mostly reported in animal husbandry waste especially from swine feedlots (Li et al. 2013). In certain strains of *Bacteroides fragilis* isolated from influent and effluent samples of WWTPs, chloramphenicol-resistant gene *catA1* was reported (Niestepski et al. 2019). As previous studies have reported the occurrence of chloramphenicol resistance genes in animal husbandry sewage and wastewater; and since none of our samples has animal waste enrichment, thus, the abundance of chloramphenicol resistance genes was low. In addition to this, results showed that beta-lactam resistance genes, beta-lactamase (*bla*), were found in pharmaceutical sediment and wetland sludge sample and were absent in the metagenome of ETP sludge sample. Previous studies have shown that expression of beta-lactamase abetted by increased efflux of beta-lactams in periplasmic space confers high-level resistance in gram negative bacteria even at low levels of beta-lactamase (Rice 2012). The occurrence of such genes highlights the risk associated with dissemination of resistance across the bacterial community in such habitats.

The trimethoprim resistance gene (*dfr*) was absent from the metagenome of the wetland sludge sample. Trimethoprim resistance genes are frequently found in the *Aeromonas* species. Also, the fish farming industry have been evidenced as reservoir of *dfr*, and they co-occur with *sul* and *int1* genes (Domínguez et al. 2019).

Strikingly, a vancomycin resistance gene were not annotated in any of three samples using the ResFinder database but was mapped against the COG database in ETP sludge and wetland sludge samples only. Such results were in congruence with previous report which states that (Li et al. 2015) the vancomycin resistance genes although present at low levels, it widely occurs in human feces, STP influent, and waste from livestock farms; however, it was absent from river water, drinking water, and sediments. Such distribution pattern of VGRs can be attributed to the fact that vancomycin is used as the last line of defense against gram-positive bacteria such as *Streptococcus pneumonia* and *Enterococcus*; it has been overused during the past last decade.

Our finding reveals that a particular environment has its own unique resistome profile which is broadly dictated by prevalence of antibiotic-stress exposed inhabiting microbial flora. The selected metagenomes of three different sites have their unique resistome pattern. However, the proximity of the ARG enrichment of ETP and pharmaceutical sediment samples can be explained by the fact that similar type of selection pressure may give rise to related distribution profile. Also, previous study (Xia et al. 2019) explained the co-occurrence of the aromatic compound degrading genes and ARGs in the microbes. The study also explains the effect of aromatic compounds on the selection and spread of ARGs. Hence, it justifies the similar richness of ARGs in ETP and pharmaceutical discharge sediment sample as depicted in the PCA plot.

The occurrence and distribution of ARGs pose a high risk by implying a chronic selection pressure on microbiota in sediments, water, and soil matrices (Walsh 2013; Yuan et al. 2019). Studies have reported the use of treatment technologies including activated carbon, UV treatment, or advanced oxidation processes for the removal of antibiotic residues in the wastewaters (Aydin et al. 2019). Another study by Du et al. (2015) demonstrated that tetracycline and sulphonamide resistance genes (*tetG*, *tetW*, *text*, and *sul1*) are found to decrease in effluent of WWTPs employing anaerobic/aerobic or membrane bioreactor (MBR) technologies. As opposed to wastewater processes, the full-scale constructed wetland systems are efficient technology for removal of *int1* and *ermK* (Lamori et al. 2019). The results of the present study also highlight the need of advancement in such water treatment technologies to reduce the risk of antibiotic resistance spread in the environment.

Conclusion

This study provides a comparative profile of antibiotic resistome across different ecological habitats. The abundant ARGs found accorded with antibiotics used amply in human and veterinary epidemics. As hypothesized, the resistome profile of the pharmaceutical sediment sample has the highest number of unique ARGs annotated. The resistance mechanism developed against tetracycline showed that ETP has a primitive level of resistance, whereas the pharmaceutical sediment has an advanced version of resistance. Among critically important resistance genes, quinolone resistance genes (*qnrS*, *qnrV*, and *qnrD*) and tetracycline resistance genes (*tetA*, *tetC*, *tetW*, *tetT*, *tetS/M*, *tetO*, and *tetM*) were found in only pharmaceutical sediment, whereas sulphonamide resistance genes (*sul1*, *sul2*, and *sul4*) were found in all the three samples with its highest amounts present in pharmaceutical sediment. The current study also paved a way towards the identification of environmental reservoirs for development, breeding, and spreading of resistant bacteria and genes. Discharge sites of

effluents from pharmaceutical industries and WWTPs are pivotal points of intervention for the mitigation of antibiotic resistance. Further studies towards an understanding of complex interaction of microbial community involved in developing antibiotic resistome in natural systems are also of prime importance. It will also help in putting limits to antibiotic levels and resistance load discharges into the environment, thereby increasing clinical lifetime of antibiotics for combating infectious diseases in humans and animals.

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Authors' contributions HJP and NAD conceptualized and designed the study; SB and VJ performed the sample collection, metagenomic DNA isolation, bioinformatics analysis, and equally contributed initial manuscript writing; NAD and AB planned the bioinformatics analysis; NAD and AB improved the manuscript.

Data availability The datasets generated and/or analyzed during the current study are available in the MG-RAST repository [<https://www.mg-rast.org/linkin.cgi?project=mgp90567>].

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval and consent to participate Not applicable.

Consent for publication Not applicable.

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